Graphic Era

MID TERM EVEN SEMESTER EXAMINATION-2026

Roll no.....

Name of the Program & Semester: B.Tech II SEM.
Name of the Course: Programming for Problem solving
Course Code: TCS201
Time: 1.5 hours

Maximum Marks: 50

Note:
(i) Answer all the questions except Que(2) by choosing any one of the sub-questions.
(ii) All 5 sub-questions in Que(2) are compulsory.
(iii) Each question carries 10 marks.

Que(1) **(10 marks)**

(a) Explain compile time and run time initialization of 2-D Arrays with an example for each using a C code. Draw a flowchart to read two binary-2-D arrays **A** and **B** of size M-rows and N-columns, where the elements are only 1s and 0s. Find & display the resultant 2-D array C by bitwise ANDing the elements of the A and B arrays.

OR

(b) Develop a code to input a square matrix of order *mXn*. Find sum of each row of matrix having odd index.

Sample Input:	**Sample Output:**
Row Size: 4 Column Size: 3 Inputted Matrix: 2 4 1 6 2 9 3 7 5 8 3 2	Sum of row index 1: 7 Sum of row index 3: 15

CO1

Que(2) Predict the output of the following programs. **(10 marks)**

1.
```
#include <stdio.h>
void main()
{
    int i, m, n, s=0;
    int mat[][3]={1,2,3,4,5,6,7,8,9};
   for(i=0;i<3;i++)
      s=s+mat[i][i];
   printf("The sum is %d", s);
}
```

2.
```
#include <stdio.h>
int main()
{
    int a[] = {10, 20, 30, 40};
    int *p = &a[3];
    int *q = &a[1];
    printf("%ld", p - q);
    return 0;
}
```

3.
```
#include <stdio.h>
int main()
{
      int a[] = {1, 2, 3};
      int *p = a;
      printf("%d %d", ++*p, *p);
      return 0;
}
```

4.
```
#include <stdio.h>
int main()
{
      char str[] = "abcdef";
      printf("%s", str + 2);
      return 0;
}
```

5.

CO1
CO3

```c
#include <stdio.h>
void modify(char *p)
{
    p = "Programming";
    printf("%s ",p);
}
int main()
{
    char str[] = "Hello";
    modify(str);
    printf("%s", str);
    return 0;
}
```

Que(3) **(10 marks)**

(a) Explain at least five string manipulation library functions supported in C along with their syntax. Also, write a C program to read a string from the user and then reverse it without using an additional string and display the modified string.

Sample Input	Inputted String: Programming is easy
Sample Output	ysae si gnimmargorP

CO1

OR

(b) Arun stores name of n employees in a single string. His boss calls him and asks him to give a count of all the names starting with the alphabet given by him. Please help Arun to write a code so that he can easily find the count.

Sample Input	Inputted String: Ajay Manan Rajat Tejus Aarya Akshit Rashi Anuj Inputted Alphabet: R
Sample Output	Total words: 2

Que(4) **(10 marks)**

(a) Draw a flowchart to pass three test marks of a student into a function and function will return sum of test marks, average test marks and grade of student.
Grade criteria:
if average >=75 then Grade = A
if average >=50 then Grade = B
Otherwise Grade = C

CO2

OR

(b) Draw a flowchart to pass a string *str* into a function named *encrypt()*. The function will encrypt the string according to following criteria. Display the final encrypted string in *main()* function.

1. All the vowels will replace by their capital case.
2. All the consonant will replace by their predecessor characters.
3. All the digits will be replaced by '#' character.

Sample Input	**Sample Output**
Inputted String: My house number is 157	Output String: Lx gOUrE mUlaEq Ir ###

Que(5) **(10 marks)**

(a) Explain the following terms with a snippet of C code wherever possible.
i. Pointer to an array ii. NULL pointer iii. Wild Pointer
iv. Dangling Pointer v. Memory Leak

CO3

OR

(b) Develop a code in C which passes a 1D array of *n* integer elements into a function *replace()* as a pointer. Function will replace all elements of array with their factorial. Print the final updated array in main() function.